Homework 8

Due: Friday, November 6 at 11:59 p.m. PT

MATH 112, Fall 2026

Instructions

- Write your full name, "Homework 8," and the date at the top of the first page.
- Show all work and explain your reasoning. Write in complete sentences.
- Typeset your solutions in LaTeX if you can.
- Submit a single PDF file to Gradescope under the assignment "Homework 8."
- Use Gradescope to electronically match problems to pages in your PDF.
- Questions? Email the instructor or come to office hours; see Moodle.
- You are strongly encouraged to work together. Write up your own solutions.

Be sure to show your work. Organize it so that another MATH 112 student who has not seen the problems can follow your reasoning from line to line. Solutions that contain only an answer, without justification or supporting work, will not receive full credit.

Problem 1 (16 points)

On any set S, the **discrete metric** is $d_{\text{disc}}(x, y) = 0$ if $x = y$ and $d_{\text{disc}}(x, y) = 1$ if $x \neq y$.

(a) Show that d_{disc} is a metric.

(b) Describe $B_{d_{\text{disc}}}(x, r)$ for $0 < r \leq 1$ and for $r > 1$.

(c) Show that every subset of S is both open and closed with respect to d_{disc}.

(d) Describe the sets $B_{d_{\text{disc}}}(x, 1)$ and $D_{d_{\text{disc}}}(x, 1)$. What do you observe?

Problem 2 (12 points)

Using the definition of the limit of a sequence, prove that

(a) $\lim_{n\to\infty} \dfrac{1}{n^3} = 0.$

(b) $\lim_{n\to\infty} \dfrac{n-1}{n+1} = 1.$

(c) $\lim_{n\to\infty} \dfrac{n^2-n}{n^2+n} = 1.$

Problem 3 (12 points)

Prove that each of the following sequences converges to 0:

$$a_n = \frac{n}{n^3+n^2+1}, \qquad b_n = \sqrt{n^2+2} - \sqrt{n^2+1}, \qquad c_n = \begin{cases} \dfrac{1}{2^n}, & \text{if } n = 1, 4, 7, 10, 13, \ldots, \\ \dfrac{1}{n^2+1}, & \text{otherwise.} \end{cases}$$

Problem 4 (10 points)

Let (a_n) be a sequence of real numbers. Show that if $\lim_{n\to\infty} a_n = a > 0$, then $a_n > 0$ for all sufficiently large n (that is, $a_n > 0$ eventually).

Problem 5 (10 points)

Let A be a nonempty subset of $\mathbb{R}$ that is upper bounded. Show that if $a = \sup A$, then there exists a sequence (a_n) of elements of A with $\lim_{n\to\infty} a_n = a$.

Hint: The ε characterization of the supremum may be helpful here.

Problem 6 (10 points)

Show that if (a_n) is a sequence of positive real numbers with $a_n \to 0$, then the set $A = \{a_n : n \in \mathbb{N}\}$ has a maximum element.